

National Taiwan University - Calculus 1 for Class 10-13 (Year 114)
Worksheet 1 : Nash equilibrium and the Intermediate Value Theorem

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Introduction.

Suppose there are two firms A and B producing a certain commodity.

- If A produces x units of commodity, the best response of B is to produce $y = f(x)$ units in order to maximize its profits. Whereas B produces y units of commodity, the best response of A is to produce $x = g(y)$ units to maximize its profits. In this context, the functions $f(x)$ and $g(y)$ are called the *best response functions* and are assumed to be *continuous*.
- A **Nash equilibrium** $(\bar{x}, \bar{y})$ is a pair of values that satisfies $\bar{y} = f(\bar{x})$ and $\bar{x} = g(\bar{y})$, or in other words, both A and B are best responding to the other's strategy.

In this worksheet, we will use the **Intermediate Value Theorem** to imply the existence of a Nash equilibrium.

Question 1.

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Suppose that two oligopolists A and B produce x and y units of a commodity respectively. It is given that :

- The price of the commodity is $p(x, y) = 92 - 0.25(x + y)$.
- The cost function for firm A is $C_A(x) = 4x + 100$.
- The cost function for firm B is $C_B(y) = 3y + 125$.
- The revenue of firm A is $R_A(x, y) = p(x, y) \cdot x$.
- The revenue of firm B is $R_B(x, y) = p(x, y) \cdot y$.
- The profit of a firm is given by its revenue minus its cost, $\pi = R - C$.

(a) Write down the profit functions, $\pi_A(x, y), \pi_B(x, y)$, for A and B.

$$\begin{aligned}\pi_A &= R_A(x, y) \cdot x - C_A(x) = [92 - 0.25(x + y)]x - (4x + 100) = -0.25x^2 + 88x - 0.25xy - 100 \\ \pi_B &= R_B(x, y) \cdot y - C_B(y) = [92 - 0.25(x + y)]y - (3y + 125) = -0.25y^2 + 89y - 0.25xy - 125\end{aligned}$$

(b) To find the best response function of A, we regard y as a constant and view the profit of A as a polynomial in x of degree 2. By completing the square, find $x_{\max}$ that maximizes the profit. Here $x_{\max}$, which depends on y , is the best response function of A. In the sequel, we will denote it as $g(y)$.

$$\begin{aligned}\pi_A(x, y) &= -0.25x^2 + (88 - 0.25y)x - 100 = -0.25 \left(x^2 - \frac{88 - 0.25y}{0.25}x \right) - 100 \\ &= -0.25 \left(x - \frac{352 - y}{2} \right)^2 \Rightarrow x_{\max} = \frac{352 - y}{2}, \quad g(y) = \frac{352 - y}{2}\end{aligned}$$

(c) Similarly, find the best response function $y_{\max} = f(x)$ of B.

$$-0.25y^2 + 89y - 0.25xy - 125 = -0.25\left(y - \frac{356-x}{2}\right)^2 + 0.25\left(\frac{356-x}{2}\right)^2 - 125$$

$$y_{\max} = \frac{356-x}{2}, \quad f(x) = \frac{356-x}{2}$$

(d) Suppose that A produces $x_1 = 80$ units. Find the best response of B which is denoted by y_1 . Then find the best response of A for y_1 which is denoted by x_2 and the best response of B for x_2 . Repeat this process and obtain $x_1, x_2, x_3 \dots$ and $y_1, y_2, y_3 \dots$. Do the sequences $x_1, x_2, x_3 \dots$ and $y_1, y_2, y_3 \dots$ seem to converge?

claim $x_1 < x_2 < x_3 \dots < x_n < \dots < y_n < \dots < y_1$

$$n=1, \quad x_1 = 80 < y_1 = 178$$

$$y_{n+1} - x_{n+1} = \frac{356 - x_{n+1}}{2} - x_{n+1} = 178 - \frac{3}{2}x_{n+1} \Rightarrow 178 - (1249 - \frac{3}{4}y_n)$$

$$\downarrow \quad \quad \quad \downarrow$$

$$\frac{332 - y_n}{2} \quad \quad \quad -71 + \frac{3}{4}y_n$$

$$y_n = \frac{356 - x_n}{2} > x_n \quad (\text{induction hypothesis})$$

$$\Rightarrow x_n < \frac{356}{3}, \quad y_n = \frac{356 - x_n}{2} > \frac{356}{3}$$

(e) Solve the system of equations $x = g(y)$ and $y = f(x)$. Verify that the solution is a Nash equilibrium.

$$x = g(y) = 176 - 0.5y \quad \text{--- ①}$$

$$y = f(x) = 178 - 0.5x \quad \text{--- ②}$$

$$\text{①} \times 2 \Rightarrow 2x = 352 - y \quad \text{--- ③}$$

$$\text{②} \Rightarrow \text{③} \Rightarrow 2x = 352 - 178 + 0.5x \Rightarrow x = 116, y = 120$$

$$\Rightarrow (\bar{x}, \bar{y}) = (116, 120)$$

For Nash Equilibrium $\bar{y} = f(\bar{x})$ & $\bar{x} = g(\bar{y})$

$$f(116) = 178 - 0.5 \cdot 116 = 120 = \bar{y}$$

$$g(120) = 176 - 0.5 \cdot 120 = 116 = \bar{x}$$

It is Nash Equilibrium

Question 2.

In general, we are given **continuous** best response functions $f(x)$ and $g(y)$.

Furthermore, suppose that there is an interval $[\alpha, \beta]$ such that

$$\alpha \leq f(x) \leq \beta \text{ and } \alpha \leq g(y) \leq \beta \text{ for all } x, y \in [\alpha, \beta]$$

i.e. the ranges of f and g on $[\alpha, \beta]$ are still contained in $[\alpha, \beta]$. We will prove that in this case there is always a Nash equilibrium $(\bar{x}, \bar{y})$ satisfying $\alpha \leq \bar{x}, \bar{y} \leq \beta$.

(a) First prove the following statement.

A **fixed point** of a function $h(x)$ is a number c in its domain such that $h(c) = c$. Use the Intermediate Value Theorem to prove that any continuous function $h(x)$ with domain $[\alpha, \beta]$ and range in $[\alpha, \beta]$ must have a fixed point. (Hint. Consider the function $H(x) = h(x) - x$. Show that $H(\alpha) \geq 0$, $H(\beta) \leq 0$ and there must be a number $c \in [\alpha, \beta]$ such that $H(c) = 0$.)

$$h[\alpha, \beta] \supset [\alpha, \beta], \quad H(x) = h(x) - x$$

domain range

$$\alpha \leq h(x) \leq \beta, \quad H(x) = h(x) - x$$

$$0 = \alpha - \alpha \leq H(\alpha) = h(\alpha) - \alpha \leq \beta - \alpha$$

$$\alpha - \beta \leq H(\beta) = h(\beta) - \beta \leq \beta - \beta = 0$$

$$1. H(\alpha) = 0 \text{ or } H(\beta) = 0, \text{ Done}$$

$$2. H(\alpha) > 0 \text{ and } H(\beta) < 0$$

$$\therefore H(\alpha) > 0 > H(\beta) \text{ and } H \text{ is continuous}$$

$$\therefore \exists c \in (\alpha, \beta) \text{ such that } H(c) = 0$$

- (b) Consider the function $h(x) = g(f(x))$. By (a), show that $h(x)$ has a *fixed point* $x_0 \in [\alpha, \beta]$, i.e. $g(f(x_0)) = x_0$. Then verify that $(\bar{x}, \bar{y}) = (x_0, f(x_0))$ is a Nash equilibrium.

$$f: [\alpha, \beta] \rightarrow [\alpha, \beta]$$

$$g: [\alpha, \beta] \rightarrow [\alpha, \beta]$$

$$h: g \circ f: [\alpha, \beta] \rightarrow [\alpha, \beta], \quad h: \text{continuous}$$

$$\text{By (a), } \exists x_0 \in [\alpha, \beta] \text{ s.t. } h(x_0) = g(f(x_0)) = x_0$$

$$\begin{bmatrix} \bar{x} = f(x_0) \\ \bar{y} = g(\bar{x}) \end{bmatrix} \quad \text{pick } y_0 = f(x_0), \text{ Then } g(y_0) = x_0$$

Therefore, (x_0, y_0) is a Nash Equilibrium